

Basic GitHub Commands – Beginner's Practical Guide

Step-by-step guide for creating a local Git repository, connecting it to GitHub, and uploading/updating a project.

1. Git vs GitHub

Git is software installed on your computer that tracks changes in your project. **GitHub** is an online platform for storing and sharing Git repositories.

2. Check Git Installation

```
git --version
```

If Git is installed, a version such as `git version 2.x.x` will be displayed.

3. Configure Git (First Time Only)

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
git config --global --list
```

4. Create and Enter a Project Folder

```
mkdir college-website
cd college-website
pwd
```

5. Initialize the Git Repository

```
git init
```

This creates the hidden `.git` folder and tells Git to start tracking the project.

6. Check Project Status

```
git status
```

Use this frequently to see which files are untracked, modified, staged, or ready to commit.

7. Add Files to the Staging Area

```
git add .
```

```
# Or add one specific file:
git add index.php
```

8. Commit Your Changes

```
git commit -m "Initial college website"
```

A commit is a saved snapshot of your project. Use a short message that explains what changed.

9. Create a Repository on GitHub

Sign in to GitHub, choose **New repository**, enter a name such as `college-website`, and create the repository. For a beginner's first upload, you can leave README, .gitignore, and license options unchecked.

10. Connect the Local Repository to GitHub

```
git remote add origin https://github.com/yourusername/college-website.git  
git remote -v
```

11. Rename the Branch to main

```
git branch -M main
```

12. Upload the Project to GitHub

```
git push -u origin main
```

After a successful push, refresh the GitHub repository page to see your files online.

13. The Regular Git Workflow

After your project is connected to GitHub, the usual workflow is:

```
Make changes to your files
↓
git status
↓
git add .
↓
git commit -m "Description of changes"
↓
git push
```

Example:

```
git status
git add .
git commit -m "Updated home page"
git push
```

14. Download an Existing GitHub Project

```
git clone https://github.com/username/project.git
cd project
```

`git clone` downloads an existing repository to your computer.

15. Get the Latest Changes

```
git pull
```

Use this when the remote GitHub repository has newer changes that you want on your computer.

16. View Commit History

```
git log
git log --oneline
```

17. See Uncommitted Changes

```
git diff
```

This shows differences between your working files and the last staged/committed state.

18. Basic Command Reference

Command	Purpose
<code>git --version</code>	Check Git version
<code>git config</code>	Configure Git
<code>git init</code>	Create a Git repository
<code>git status</code>	Check project status
<code>git add .</code>	Stage all changes
<code>git add filename</code>	Stage one file
<code>git commit -m "message"</code>	Save a snapshot
<code>git push</code>	Upload changes to GitHub

Command	Purpose
<code>git pull</code>	Download latest changes
<code>git clone</code>	Download a repository
<code>git log</code>	View commit history
<code>git branch</code>	View branches
<code>git branch -M main</code>	Rename branch to main
<code>git remote -v</code>	View remote repository
<code>git diff</code>	View changes

19. Common Error: 'not a git repository'

`fatal: not a git repository (or any of the parent directories): .git`

This usually means you are running a Git command outside your project folder, or the folder has not been initialized. Enter the correct project folder and run `git init` if necessary.

```
cd D:\projects\college-website
git status
```

```
# If it is a new project:
git init
```

20. Five Commands to Memorize First

```
git status
git add .
git commit -m "My changes"
git push
git pull
```

Easy memory: Check → Add → Save → Upload → Download

21. Complete First-Time Example

```
cd D:\projects\college-website
git init
git status
git add .
git commit -m "Initial college website"
git remote add origin https://github.com/yourusername/college-website.git
git branch -M main
git push -u origin main
```

Tip: If GitHub asks for authentication, follow GitHub's current sign-in/authentication process. Do not share your password or access token with anyone.